LEARNER GUIDE

Substituting positive, negative and fractional values

Formulae and Rearranging · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

To **substitute** is to replace each letter with its value. With negative numbers and fractions, two habits keep you accurate. First, write each value **in brackets**: if $x = -3$, then x^2 becomes $(-3)^2$. Second, follow the **order of operations**: brackets, then powers, then × and ÷, then + and –. A power belongs only to what it is attached to, so $(-3)^2 = 9$ but $-3^2 = -9$.

What you are learning to notice

- Write every substituted value in brackets, especially negatives: 4 × (–2), not 4 × –2.
- Two negative numbers multiplied together give a positive answer: $(-3)^2 = (-3) \times (-3) = 9$.
- Subtracting a negative number is the same as adding: 10 – (–6) = 10 + 6.
- Fractions follow the same rules: if a = ½, then 2a = 2 × ½ = 1.

Moving through it, one step at a time

Copy the expression with every letter replaced by its value in brackets. Then work through it one step at a time in the order of operations, writing each stage on a new line. Finally, ask whether the answer makes sense — especially in a real formula.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A negative value with a square

Work out $3x^2 - 2x$ when $x = -2$.

1. Substitute, in brackets.	$3 \times (-2)^2 - 2 \times (-2)$
2. Powers first: $(-2)^2 = 4$.	$3 \times 4 - 2 \times (-2)$
3. Then multiplication.	$12 - (-4)$
4. Subtracting a negative is adding.	$12 + 4 = 16$

$3x^2 - 2x = 16$ when $x = -2$.

Example 2 — A real formula with a negative value

Use F = 1.8C + 32 to convert –10 °C to °F.

1. Substitute C = –10, in brackets.	F = 1.8 × (–10) + 32
2. Multiplication first.	F = –18 + 32
3. Then addition.	F = 14
4. Does it make sense? Freezing is 32 °F.	14 °F is below freezing ✓

–10 °C is 14 °F.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (calculating with negative numbers, and the notation from Lesson 1), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.